Joint Models for Marker and Failure Data under Dual Censoring Schemes

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Overview

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3. Efficiency with a Correctly Specified Joint Model
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Motivation

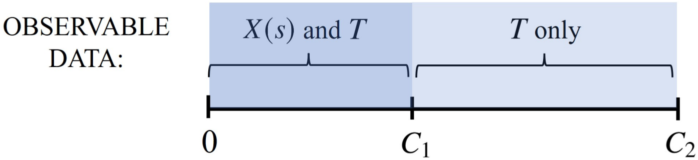
Dual Censoring

Dual censoring occurs when life history processes are censored at two different times.

Boruvka and Cook (2016)

This is common in studies involving longitudinal and failure processes:

- Longitudinal ("marker") data $\{X(s), s > 0\}$ collected over $(0, C_1]$
- Failure data T collected over $(0, C_2]$, $C_2 > C_1$



$\Longrightarrow$ missing marker data over $(C_1, C_2]$

Illustrative Settings

RCT on prostate cancer metastasis to bone: Saad et al. (2004)

- A biochemical marker of excess bone formation, bone-specific alkaline phosphatase, is measured every 3 months for 2 years
- Skeletal-related events and death are recorded up to 4 years

Finnish smoking study: Gutiérrez-Torres et al. (2024)

- Smoking behaviour is recorded every 4 months for 8 years
- Lung cancer diagnoses are identified from cancer registry up to 27 years

Past Work – Multiple Imputation

One option for accommodating the missing marker data is *multiple imputation (MI)*.

MI for failure time models has been investigated:

- Some methods yield estimates that are only approximately valid (even asymptotically), which may lead to bias White and Royston (2009); Murad et al. (2020)
- Substantive model compatible (SMC) imputation avoids approximation, but imputing time-varying covariates is still challenging Bartlett et al. (2015)
  - Two-fold fully conditional specification (FCS) avoids numerical issues that may arise with many time points, high serial correlation, ... Nevalainen et al. (2009); Welch et al. (2014)
    ... but it has had mixed performance in simulations Silva et al. (2017); Huque et al. (2018)

Combining SMC and two-fold FCS could be an interesting line of investigation.

Past Work – Joint Modeling

Another option is to *jointly model* the marker and failure processes.

Types of joint models:

- Shared/correlated random effect models .... Rizopoulos (2012)
  - R software: `joineR`, `JM`, `JMbayes`, `frailtypack`, etc.
- Joint latent class models .... Proust-Lima et al. (2014)
  - R software: `lcmm`
- Multivariate copula joint models .... Zhang et al. (2026)
- Multistate models .... Cook and Lawless (2018)
  - R software: `survival`, `msm`

Multistate models tend to be robust for estimates of state occupancy (Aalen et al., 2001), but these usually require the same length of follow-up for the marker and failure processes.

Present Work

Goal: Investigate strategies for studying the relationship between a marker $X(s)$ and failure time T under a dual censoring scheme.

Approach: Formulate a joint multistate model for $X(s)$, T, and the random censoring process, then use an EM algorithm to accommodate the missing marker data between C_1 and C_2.

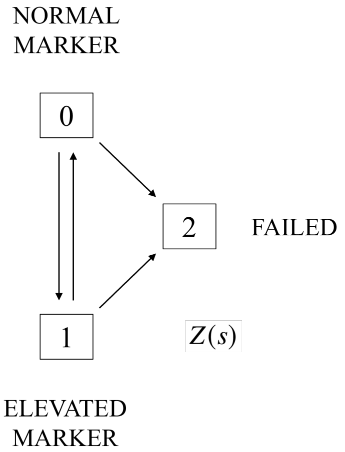
To focus on the dual censoring issue, we consider a simplified setting:

- Discrete time
- Binary marker $X(s)$
- Failure time T and random censoring times C_1, C_2 all mutually independent

Joint Model for Marker, Failure, and Random Censoring Processes

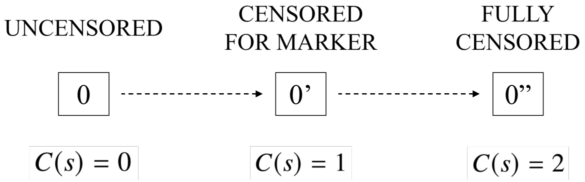
Notation

Marker-failure process:

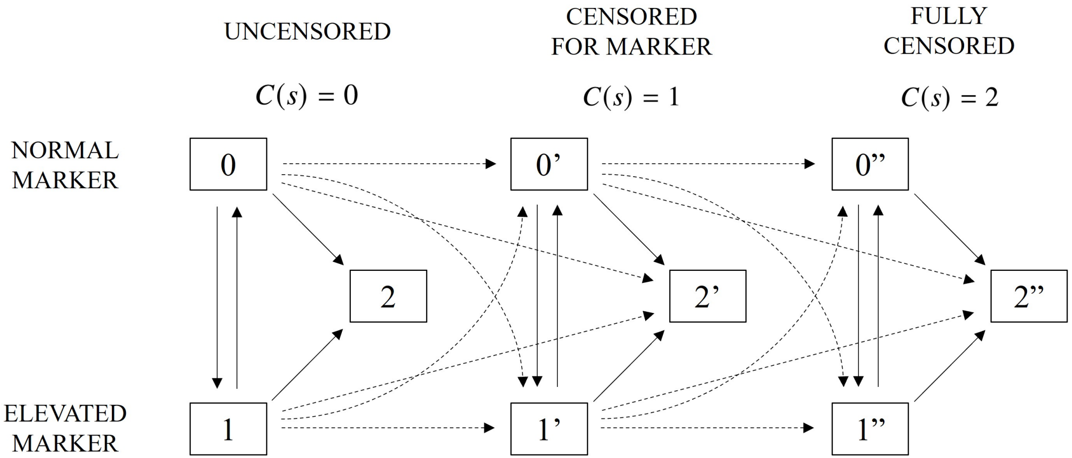


- $N(s) = 1$ if $T \leq s$ (failed by time s)
  0 otherwise
- $X(s) = 1$ if marker is elevated and $N(s) = 0$
  0 if marker is normal and $N(s) = 0$

Random censoring process:



Joint Multistate Diagram



Discrete-Time Intensity Models

Let V_i be a time-fixed binary covariate. For $z_i, v_i \in \{0, 1\}$:

Intensity for failure:

$$\begin{aligned}\mathbb{P}\big(Z_i(s) = 2 \,\big|\, Z_i(s-1) = z_i, C_i(s-1), V_i = v_i\big) \\ = \operatorname{expit}\left\{\beta_0(s) + \beta_1 z_i + \beta_2 v_i + \beta_3 z_i v_i\right\} = \pi_i^N(s; \beta)\end{aligned} \tag{1}$$

Intensity for elevated marker:

$$\begin{aligned}\mathbb{P}\big(Z_i(s) = 1 \,\big|\, Z_i(s) \neq 2, Z_i(s-1) = z_i, C_i(s-1), V_i = v_i\big) \\ = \operatorname{expit}\left\{\alpha_0(s) + \alpha_1(s) z_i + \alpha_2 v_i + \alpha_3 z_i v_i\right\} = \pi_i^X(s; \alpha)\end{aligned} \tag{2}$$

Random censoring intensities: (independent and non-informative)

$$\begin{aligned}\mathbb{P}\big(C_i(s) = 1 \,\big|\, Z_i(s-1), C_i(s-1) = 0, V_i = v_i\big) = \operatorname{expit}(\gamma_1) \\ \mathbb{P}\big(C_i(s) = 2 \,\big|\, Z_i(s-1), C_i(s-1) = 1, V_i = v_i\big) = \operatorname{expit}(\gamma_2)\end{aligned} \tag{3}$$

Likelihood Construction

Let τ be an administrative censoring time.

The complete data likelihood corresponds to no early marker censoring:

$$\mathcal{L}_i(\alpha,\beta) \propto \prod_{s=1}^{\tau} \Big\{ \pi_i^N(s;\beta)^{\mathbb{I}(Z_i(s)=2)} \left[1-\pi_i^N(s;\beta)\right]^{\mathbb{I}(Z_i(s)\neq 2)} \cdot \pi_i^X(s;\alpha)^{\mathbb{I}(Z_i(s)=1)} \left[1-\pi_i^X(s;\alpha)\right]^{\mathbb{I}(Z_i(s)=0)} \Big\}^{\mathbb{I}(Z_i(s-1)\neq 2)} \tag{4}$$

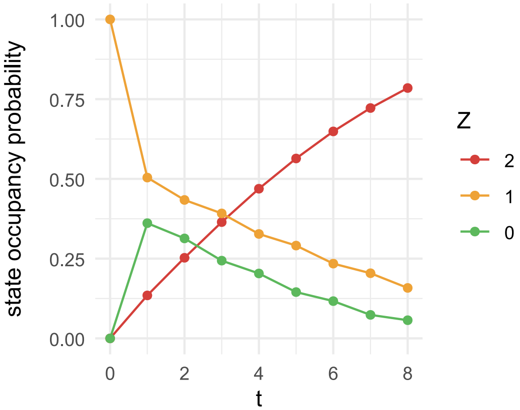
In the observed data, we do not know $Z_i(s) = 0$ vs. 1 when $C_i(s) = 1$.

We use an EM algorithm: Dempster et al. (1977)

- Weighted logistic regression with a pseudo-subject for each possible sequence of $Z_i(s) \in \{0, 1\}$, each weighted by its conditional probability given the observed data
- Variance estimation via Louis (1982)

Efficiency with a Correctly Specified Joint Model

Parameter Settings – Example

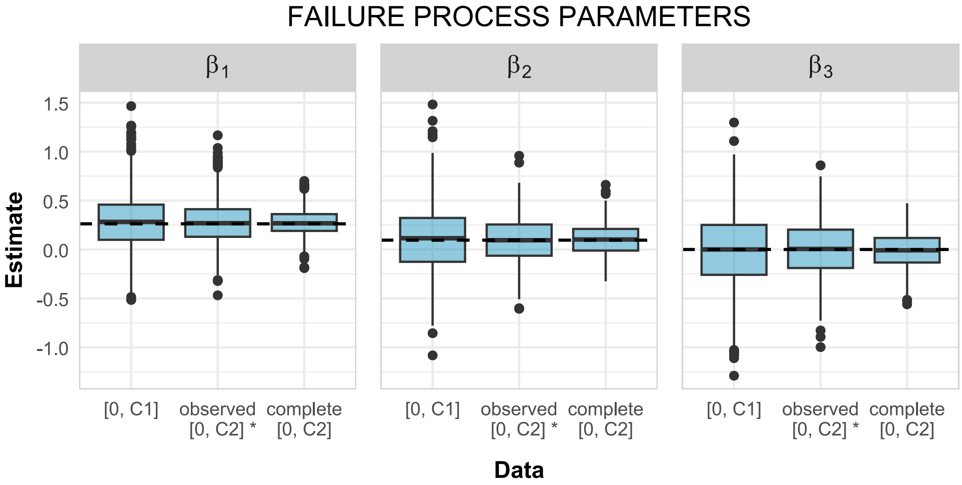


Administrative censoring at $\tau = 8$.

% expected marker data missing:

$$\mu = \frac{\mathbb{E}\{\#\text{ visits with } C(s) = 1\}}{\mathbb{E}\{\#\text{ visits with } C(s) \in \{0,1\}\}}$$
$$= \frac{\sum_{s=1}^{\tau} \mathbb{P}(C(s) = 1)}{\sum_{s=1}^{\tau} \mathbb{P}(C(s) < 2)} = 0.4$$

Simulation Results



* uses proposed method

Sensitivity to Misspecification of the Marker Model

True & Fitted Models

We now consider misspecification of the marker model

$$\mathbb{P}\big(Z_i(s) = 1 \,\big|\, Z_i(s) \neq 2, Z_i(s-1) = z_i, V_i = v_i\big) = \pi_i^X(s;\alpha)$$

Time non-homogeneous process:

True model: $\pi_i^X(s;\alpha) = \operatorname{expit}\left\{\alpha_0(s) + \alpha_1(s) z_i + \alpha_2 v_i + \alpha_3 z_i v_i\right\}$

Fitted model: $\pi_i^X(s;\alpha) = \operatorname{expit}\left\{\alpha_0 + \alpha_1 z_i + \alpha_2 v_i + \alpha_3 z_i v_i\right\}$

Omitted covariate:

True: $\pi_i^X(s;\alpha) = \operatorname{expit}\left\{\alpha_0(s) + \alpha_1(s) z_i + \alpha_2 v_i + \alpha_3 z_i v_i + \alpha_4 v_i^* + \alpha_5 z_i v_i^*\right\}$

Fitted: $\pi_i^X(s;\alpha) = \operatorname{expit}\left\{\alpha_0(s) + \alpha_1(s) z_i + \alpha_2 v_i + \alpha_3 z_i v_i\right\}$

Heterogeneous Markov processes:

True: $\pi_i^X(s \mid u_i;\alpha) = 1 - \exp\left\{-\exp\{\alpha_0 + \alpha_2 v_i + u_i\}\right\}$ for random effect u_i

Fitted: $\pi_i^X(s;\alpha) = \operatorname{expit}\left\{\alpha_0(s) + \alpha_1(s) z_i + \alpha_2 v_i + \alpha_3 z_i v_i\right\}$

Effects of Misspecification on Inference

Let $\tilde{\vartheta}$ denote the estimator of $\theta = (\alpha', \beta')'$ under a given misspecified fitted model. Then

$$\sqrt{n}(\tilde{\vartheta} - \vartheta^*) \xrightarrow{D} MVN(0, \Sigma)$$

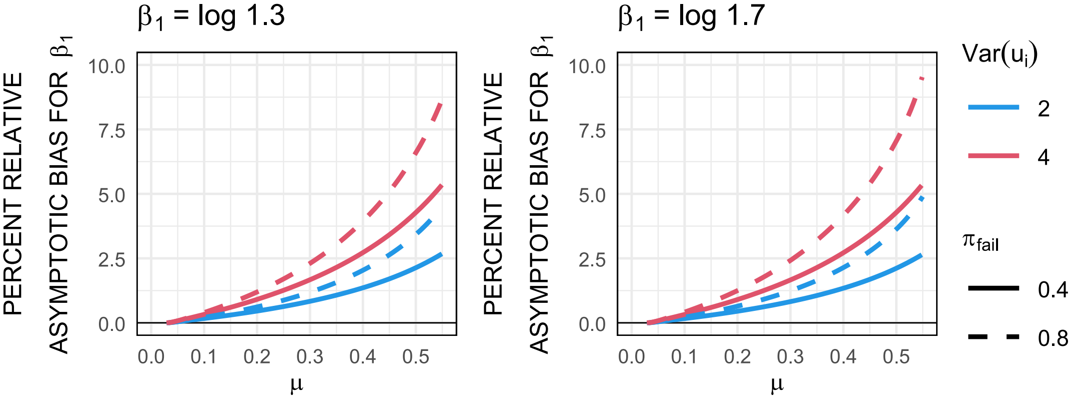
White (1982)

where Σ is calculated from the misspecified score function using a sandwich form.

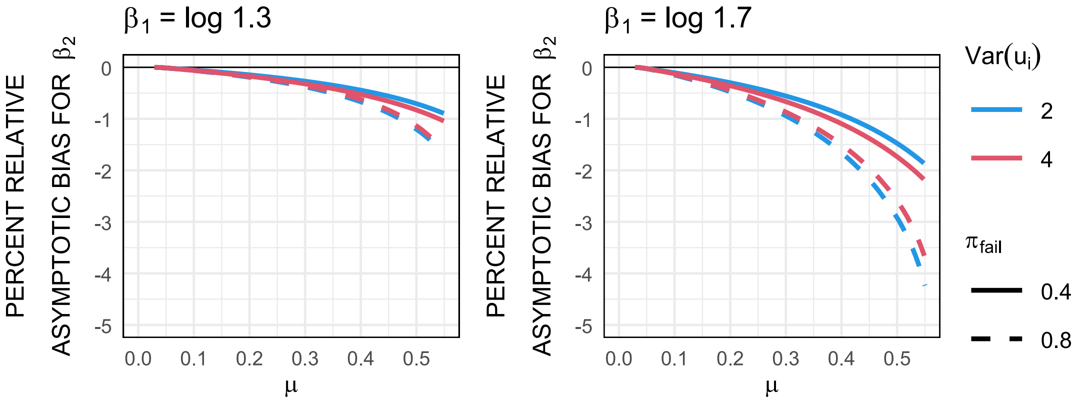
We calculated ϑ^* and Σ for a variety of parameter settings in each of the three scenarios.

The following slides show example results from the *heterogeneous Markov process* scenario, for which the magnitude of the percent relative asymptotic bias was the greatest.

Asymptotic Results – Heterogeneous Markov Process



Asymptotic Results – Heterogeneous Markov Process



Discussion

Discussion

Key takeaways:

- Using the proposed multistate approach with a correctly specified model improves efficiency substantially when estimating parameters for the failure process
- Estimation and inference about the failure process are quite robust to the marker model misspecifications considered here

Extensions in progress & future work:

- Continuous time
- More complicated latent multistate disease processes (e.g. add competing risks)
- More complicated observation processes (e.g. intermittent observations, different censoring times for competing risks)

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Thank you!

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